

Appendix S1

Detecting disease progression from animal movement using hidden Markov models

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Ecology

Section S1. Data Summary

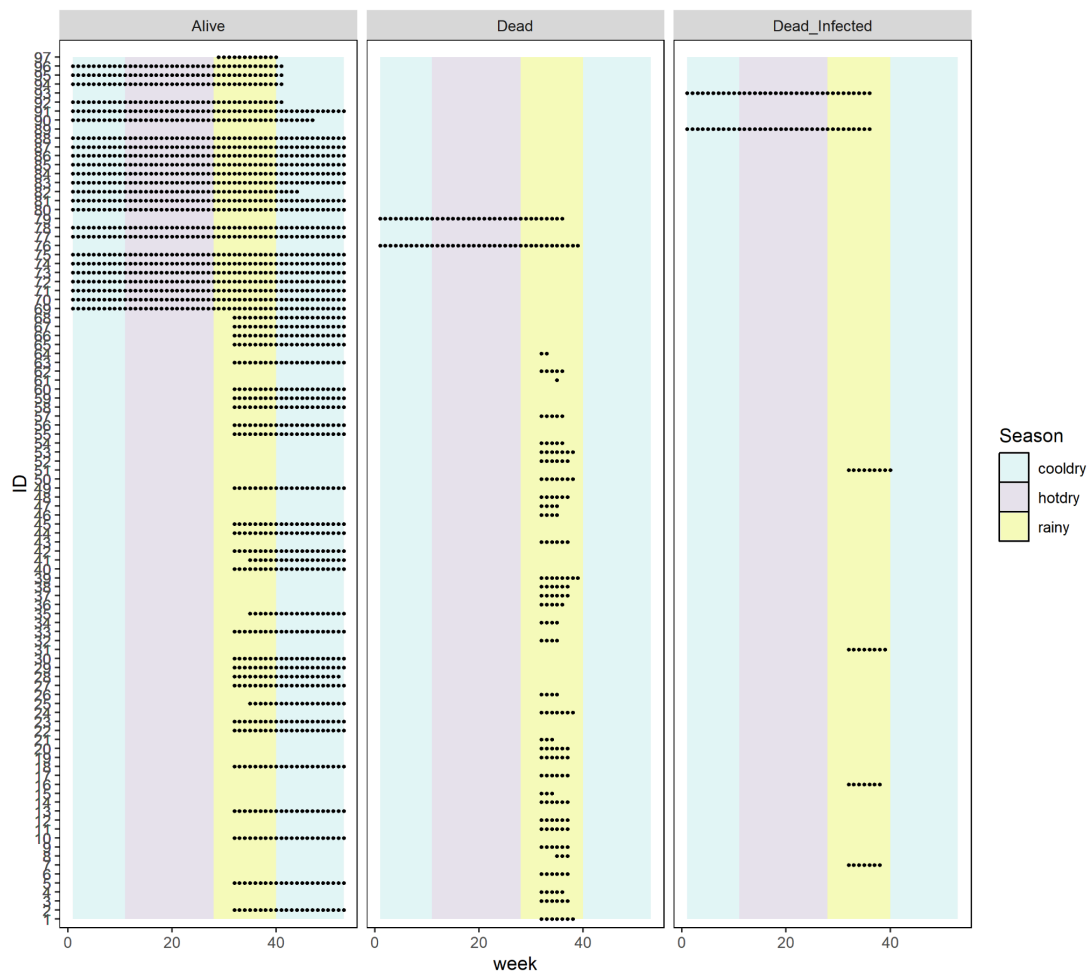


Figure S1. Summary of Oryx movement data availability. The study population of 84 tracked oryx included 46 susceptible alive, 32 susceptible (dead), and 6 known infected (dead infected) individuals. Black dots indicate the availability of movement data for a given oryx during a given week of 2018. Colored boxes indicate the three seasons characteristic of central Chad. Our study period included only the 2018 rainy season, when the majority of released oryx (Release Group 4) died within two months of release.

Table S1. Summary of study population. Here we present disease classification (Alive, Dead presumed infected, and Dead confirmed infected), release group, date of arrival to the RFOROA, date each individual was fit with a tracking device, date of release into the reserve, and date of death (based on alerts from tracking devices and field observations) for the 84 oryx in our study population.

ID	Disease classification	Release group	Release site arrival date	Date collared	Release date	Death date
1	Dead	4	02/12/2018	07/22/2018	08/06/2018	9/20/2018
2	Alive	4	02/19/2018	07/22/2018	08/06/2018	6/10/2019
3	Alive	4	02/12/2018	07/21/2018	08/06/2018	NA
4	Dead	4	02/16/2018	07/21/2018	08/06/2018	9/15/2018
5	Dead_Infected	4	02/12/2018	07/22/2018	08/06/2018	9/24/2018
6	Dead	4	02/19/2018	07/21/2018	08/27/2018	9/12/2018
7	Dead	4	02/12/2018	07/21/2018	08/06/2018	9/16/2018
8	Alive	4	02/16/2018	07/22/2018	08/06/2018	2/25/2020
9	Dead	4	02/19/2018	7/22/2018	8/6/2018	9/15/2018
10	Dead	4	02/16/2018	07/22/2018	08/06/2018	9/13/2018
11	Alive	4	02/12/2018	07/22/2018	08/06/2018	NA
12	Dead	4	02/16/2018	07/21/2018	08/06/2018	9/16/2018
13	Dead	4	02/12/2018	07/21/2018	08/06/2018	8/26/2018
14	Dead_Infected	4	02/16/2018	07/22/2018	08/06/2018	9/18/2018
15	Dead	4	02/19/2018	07/22/2018	08/06/2018	9/25/2018
16	Alive	4	02/16/2018	07/22/2018	08/06/2018	NA
17	Dead	4	02/12/2018	07/21/2018	08/06/2018	9/13/2018
18	Dead	4	02/19/2018	07/21/2018	08/06/2018	9/15/2018
19	Dead	4	02/16/2018	07/22/2018	08/06/2018	8/22/2018
20	Alive	4	02/16/2018	07/21/2018	08/06/2018	NA
21	Alive	4	02/12/2018	07/21/2018	08/06/2018	NA

22	Dead	4	02/19/2018	07/21/2018	08/06/2018	9/21/2018
23	Alive	4	02/19/2018	07/21/2018	09/01/2018	NA
24	Dead	4	02/19/2018	07/21/2018	08/06/2018	9/2/2018
25	Alive	4	02/12/2018	07/21/2018	08/06/2018	NA
26	Alive	4	02/16/2018	07/21/2018	08/06/2018	NA
27	Alive	4	02/12/2018	07/22/2018	08/06/2018	NA
28	Alive	4	02/16/2018	07/21/2018	08/06/2018	NA
29	Dead_Infected	4	02/19/2018	07/21/2018	08/06/2018	9/29/2018
30	Dead	4	02/19/2018	07/22/2018	08/06/2018	8/30/2018
31	Alive	4	02/19/2018	07/21/2018	08/06/2018	NA
32	Dead	4	02/16/2018	07/21/2018	08/06/2018	8/27/2018
33	Alive	4	02/19/2018	07/21/2018	09/02/2018	NA
34	Dead	4	02/12/2018	07/22/2018	08/06/2018	9/3/2018
35	Dead	4	02/12/2018	07/22/2018	08/06/2018	9/12/2018
36	Dead	4	02/16/2018	07/21/2018	08/06/2018	9/11/2018
37	Dead	4	02/19/2018	07/22/2018	08/06/2018	9/24/2018
38	Alive	4	02/16/2018	07/21/2018	08/06/2018	NA
39	Alive	4	02/16/2018	07/22/2018	09/01/2018	NA
40	Alive	4	02/16/2018	07/21/2018	08/06/2018	6/24/2020
41	Dead	4	02/19/2018	07/22/2018	08/06/2018	9/11/2018
42	Alive	4	02/16/2018	07/22/2018	08/06/2018	NA
43	Alive	4	02/12/2018	07/22/2018	08/06/2018	NA
44	Dead	4	02/16/2018	07/22/2018	08/06/2018	8/29/2018
45	Dead	4	02/12/2018	07/21/2018	08/06/2018	8/30/2018
46	Dead	4	02/16/2018	07/21/2018	08/06/2018	9/14/2018
47	Alive	4	02/12/2018	07/22/2018	08/06/2018	NA
48	Dead	4	02/12/2018	07/21/2018	08/06/2018	9/21/2018

49	Dead_Infected	4	02/12/2018	07/22/2018	08/06/2018	10/2/2018
50	Dead	4	02/16/2018	07/22/2018	08/06/2018	9/14/2018
51	Dead	4	02/12/2018	07/21/2018	08/06/2018	9/19/2018
52	Dead	4	02/19/2018	07/22/2018	08/06/2018	9/7/2018
53	Alive	4	02/19/2018	07/22/2018	08/06/2018	NA
54	Alive	4	02/16/2018	07/22/2018	08/06/2018	NA
55	Dead	4	02/19/2018	07/22/2018	08/06/2018	9/8/2018
56	Alive	4	02/16/2018	07/22/2018	08/06/2018	NA
57	Alive	4	02/16/2018	07/22/2018	08/06/2018	NA
58	Alive	4	02/19/2018	07/21/2018	08/06/2018	NA
59	Dead	4	02/12/2018	07/21/2018	08/28/2018	9/1/2018
60	Dead	4	02/12/2018	07/22/2018	08/06/2018	9/4/2018
61	Alive	4	02/12/2018	07/22/2018	08/06/2018	NA
62	Dead	4	02/16/2018	07/21/2018	08/06/2018	8/21/2018
63	Alive	4	02/16/2018	07/22/2018	08/06/2018	NA
64	Alive	4	02/19/2018	07/21/2018	08/06/2018	NA
65	Alive	4	02/12/2018	07/22/2018	08/06/2018	NA
66	Alive	4	02/19/2018	7/22/2018	8/6/2018	NA
67	Alive	3	01/19/2017	7/31/2017	9/25/2017	3/2/2021
68	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
69	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
70	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
71	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
72	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
73	Alive	3	11/16/2016	10/11/2016	8/3/2017	10/31/2018
74	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
75	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA

76	Alive	3	01/19/2017	7/31/2017	8/3/2017	NA
77	Alive	3	11/16/2016	7/31/2017	8/3/2017	NA
78	Dead_Infected	3	01/19/2017	7/31/2017	8/3/2017	9/7/2018
79	Alive	2	11/16/2016	10/11/2016	1/21/2017	NA
80	Dead_Infected	2	11/16/2016	10/11/2016	1/21/2017	9/4/2018
81	Alive	2	11/16/2016	10/11/2016	1/21/2017	NA
82	Alive	2	11/16/2016	10/11/2016	1/21/2017	NA
83	Alive	2	11/16/2016	10/11/2016	1/21/2017	NA
84	Alive	2	11/16/2016	10/11/2016	1/21/2017	NA

Table S2. Summary of the six infected dead individuals. Most of the individuals were co-infected from several pathogen and parasite diseases such as Rift Valley Fever (RVF), Peste des Petits Ruminants (PPR), Hemorrhagic septicemia, Pasteurellosis, and Babesiosis. Some were found dead and underwent an autopsy, while two individuals (29 and 49) were recaptured for further testing while alive.

ID	Testing methods	Infections
5	Autopsy	RVF, PPR, Hemorrhagic septicemia
14	Autopsy	Hemorrhagic septicemia
29	Recapture and vet tests (PCR)	RVF, Babesiosis
49	Recapture and vet tests (PCR)	RVF, Babesiosis
78	Autopsy	PPR, Hemorrhagic septicemia
80	Autopsy	Pasteurellosis, RVF, PPR, Babesiosis, Hemorrhagic septicemia

Section S4. Time series of mean daily step length.

We plotted mean daily step length versus time since release for all oryx in the study population (up to 90 days). We resampled oryx movement data to hourly locations to create a series of steps with a constant sampling interval and calculated the mean step length per day.

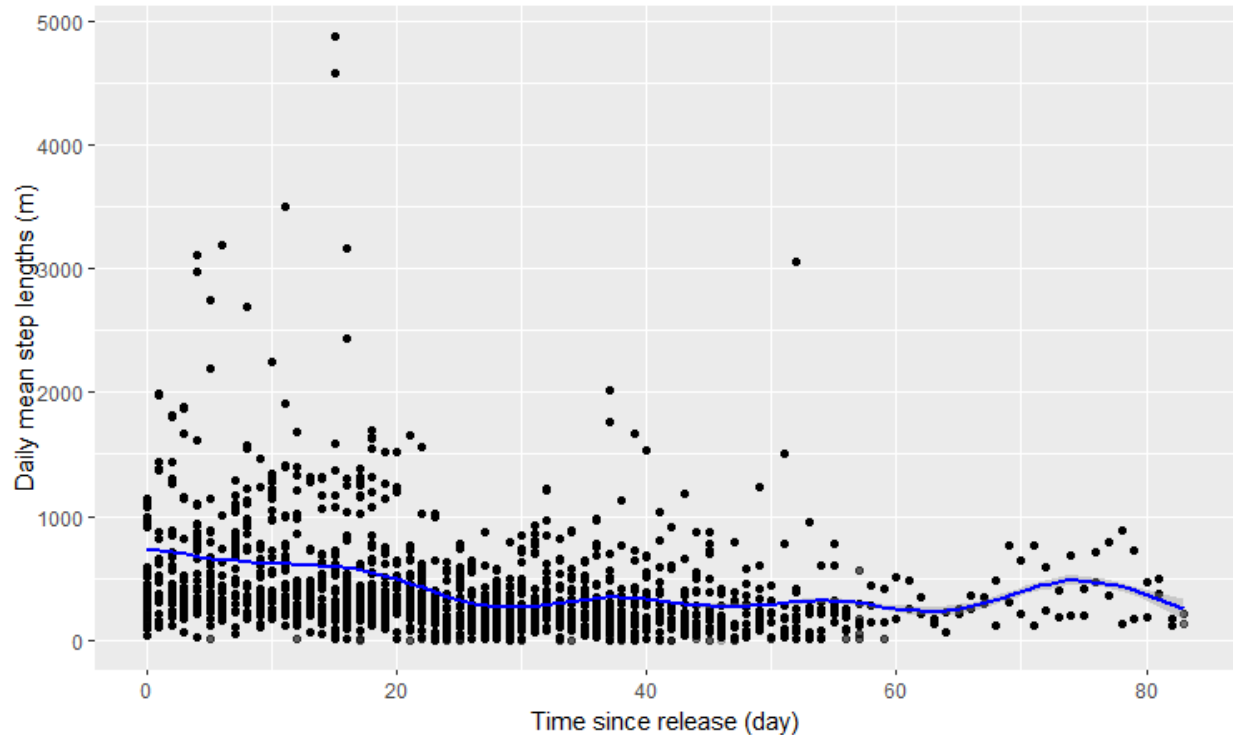


Figure S2. Time series of daily mean step lengths since the time release from 84 tagged individuals. The blue line indicates the smooth line of the total daily mean step lengths across the tagged individuals.

References

Mertes, K., Stabach, J.A., Songer, M., Wachter, T., Newby, J., Chuven, J., Al Dhaheri, S., Leimgruber, P. and Monfort, S., 2019. Management background and release conditions structure post-release movements in reintroduced ungulates. *Frontiers in ecology and Evolution*, 7, p.470.